

Chapter 28 Magnetic Fields

Magnetism

Poles of a magnet are the ends where objects are **most strongly attracted**. (Two poles North and south)

Like poles repel each other, opposite magnetic poles attract each other. (similar to electric charges)

Magnetic poles **cannot be isolated**.

Magnetic Field

The strength of a magnetic field is represented by **magnetic field strength**, a physical quantity that reflects the **characteristics of the magnetic force**.

Direction is indicated by **the north pole of a compass needle**. Magnitude can be given by **the force exerted on a moving charge**. (Called **Lorentz force**)

$$\vec{F} = q\vec{v} \times \vec{B}$$

(SI unit T , $1T = 10^4 G$).

The forces exerted on a charge in an electric and magnetic field is

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

The direction of Lorentz force can be determined by **right-hand** rule (cross product).

Note for **negative charge**.

The Lorentz force is **always** perpendicular to $\vec{v}$ and $\vec{B}$.

Magnetic Field Lines and Magnetic Flux

A series of directional curves are drawn such that **the tangent at each point** on the curve **represents the direction of the magnetic field strength** at that point, and the **density** of the curves is **proportional to the magnitude of the magnetic field strength**.

Characteristics:

- do not intersect
- closed curves without a beginning or end

Magnetic Flux:

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

(Unit: Wb)

The principle of magnetic flux continuity (Gauss's Law of magnetic fields)

$$\oint \vec{B} \cdot d\vec{A} = 0$$

That is, no magnetic core or magnetic monopole.

Motion of Charged Particles in Magnetic Field

Uniform magnetic field

- $\vec{v} \parallel \vec{B}$, $\vec{F} = 0$. Uniform linear motion.
- $\vec{v} \perp \vec{B}$, $F = qvB = \frac{mv^2}{R}$. Uniform circular motion.

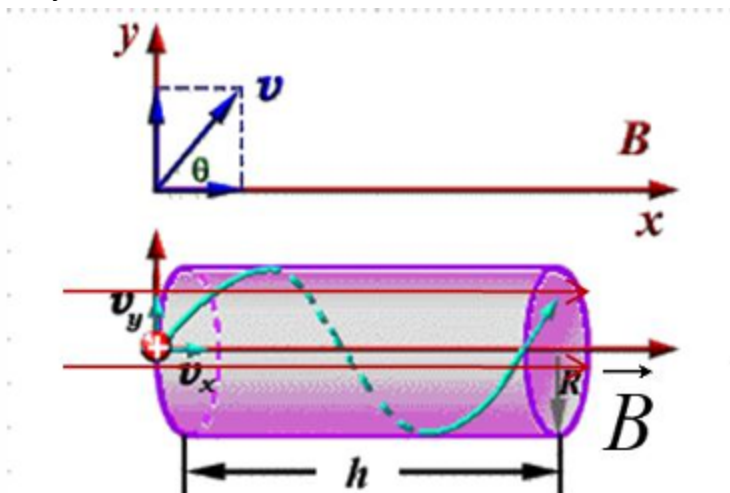
Thus, cyclone radius and cyclone period:

$$R = \frac{mv}{qB}$$

$$T = \frac{2\pi R}{v} = \frac{2\pi m}{qB}$$

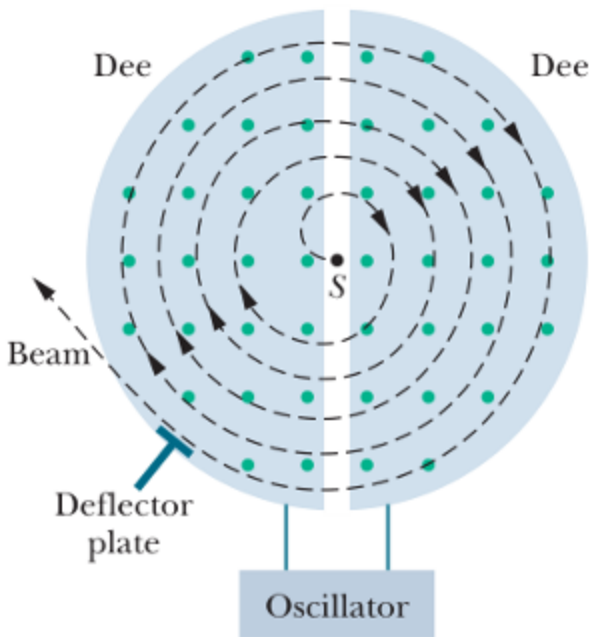
The period is independent of both speed and radius.

- $\vec{v}$ and $\vec{B}$ has angle θ
 $\vec{v} = \vec{v}_\perp + \vec{v}_\parallel$
 where $v_\perp = v \sin \theta$, $F_m = qv_\perp B$, uniform circular motion
 $v_\parallel = v \cos \theta$, $F_m = 0$, uniform linear motion.
 Totally, Helical Path.



Applications of Motion of Charged Particles

Cyclotron



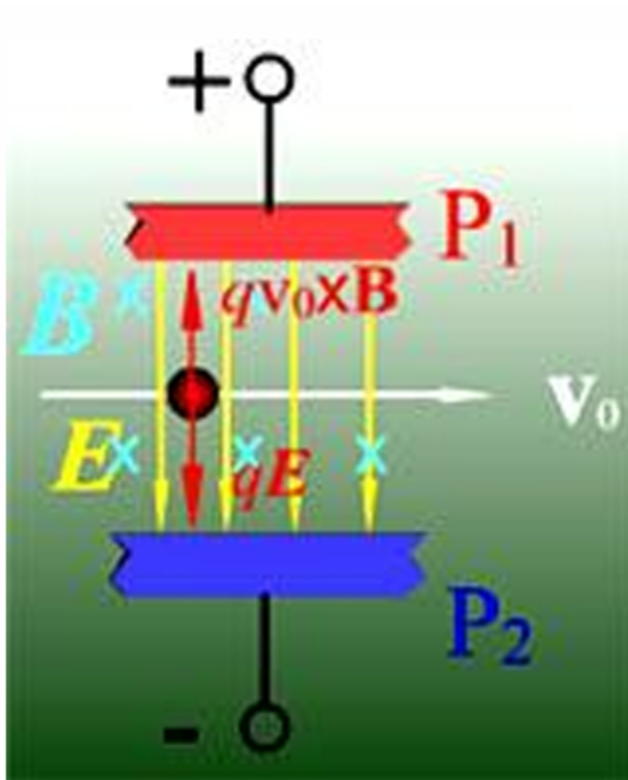
Consist of two D-shaped electrode. Electrode is in the uniform magnetic fields and inside two dee is alternating electric field. Particles injected in S and then accelerate into electrode.

Uniform circular motion inside the Dee, accelerating in the electric field.

$$R = \frac{mv}{qB}$$
$$\omega = \frac{qB}{m}$$

where angular speed is independent to velocity to ensure the frequency same to the alternating electric field.

Velocity Selector



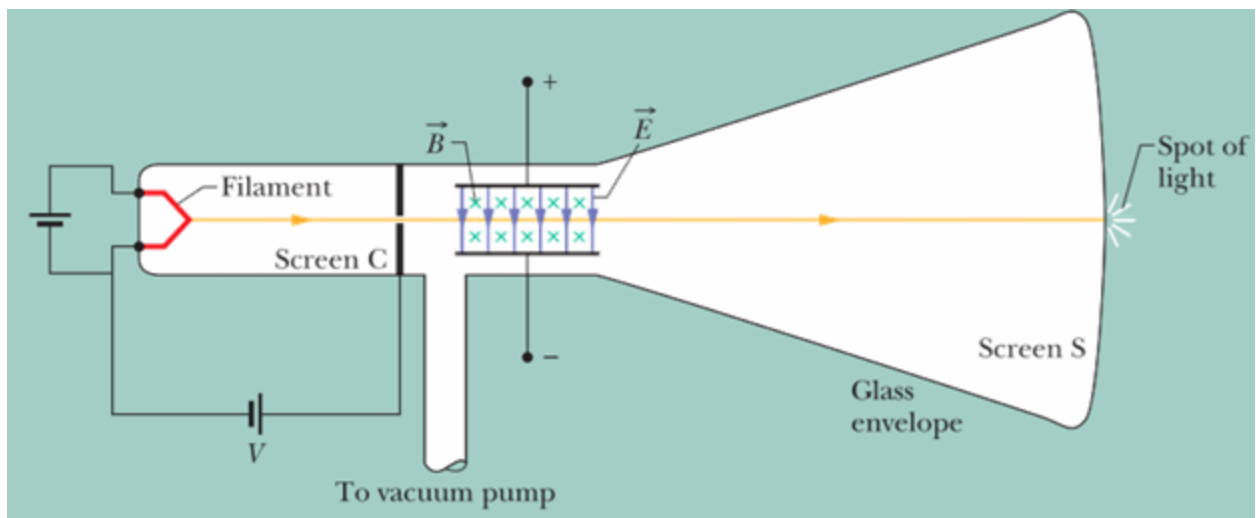
When a charged particle enters perpendicularly into a uniform electric field and a uniform magnetic field that are orthogonal to each other, the velocity of the particle passing through without deflection satisfies the following condition:

$$qvB = Eq$$

that is,

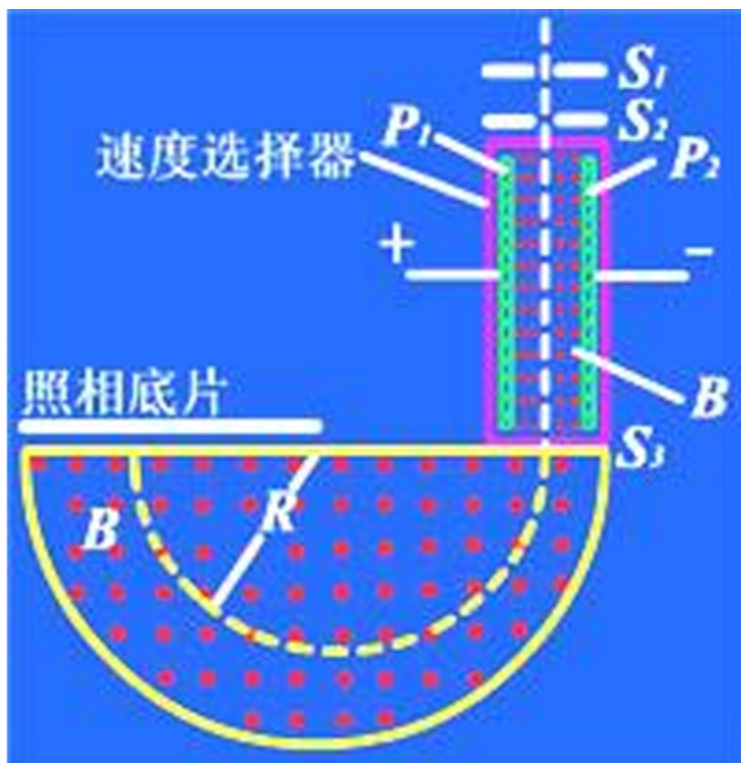
$$v = \frac{E}{B}$$

Thomson's e/m Experiment (Discovery of the Electron)



Use magnetic field and electric field discover the electron. (e/m is always definite value)

Mass Spectrometers



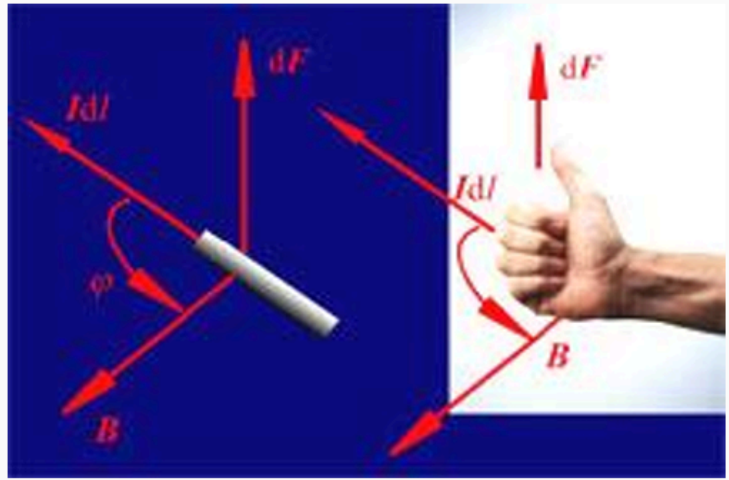
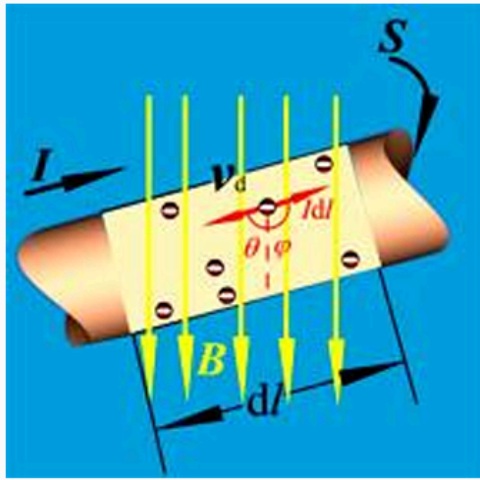
The mass m of the particle can be determined by measuring R .

$$v = \frac{E}{B}$$

$$R = \frac{mv}{qB}$$

$$m = \frac{qBR}{E}$$

Magnetic Force on a Current-Carrying Wire



$$\vec{F} = \oint I d\vec{l} \times \vec{B}$$

$$(\vec{v}_d q \times \vec{B} = \vec{v}_d q n S d\vec{l} \times \vec{B} = \vec{J} S d\vec{l} \times \vec{B} = I d\vec{l} \times \vec{B})$$

In essence, it is the macroscopic expression of **Lorentz force**.